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Institute for International Economic Studies
Stockholm University
FAO: Viktoria Garvare
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Dear Sir or Madam,

Greetings! My name is José-Elías Gallegos and I am a first-year PhD student with advanced standing at the Stockholm School of Economics. I would very much like to have the chance to write my thesis at IIES, participating and taking advantage of the outstanding faculty at the institute.

I am aware that the Graduate Position is typically oriented towards students in their second year, when students are beginning to conduct research. In this regard, although I am currently in my first year, I have already completed the first-year coursework - having been waived from several classes due to PhD coursework previously taken at Universidad Carlos III. Among the three first-year courses completed, however, I have consistently performed near the top of the cohort among students in the Stockholm Doctoral Program in Economics, including earning the highest overall mark in the *Macroeconomics I* course, taught by Timo Boppart and Lars Ljungqvist.

Among the second-year courses I have taken this year, I have also performed very well, including earning "Pass with Distinction" in the second-year macroeconomics course, *Quantitative Macroeconomic Methods* (taught by Kathrin Schlafman and Kurt Mitman). I am also currently taking Monetary Economics (taught by Johan Söderberg, Andreas Westermark and Martin Flodén). Hence, after this year I will only need to take 3 more courses.

Having had the opportunity to take second-year coursework, I will begin my research at the conclusion of this term. My primary research interests include the Economics of Ageing and Social Security as well as Monetary Economics. Among my current research agenda, there are several ideas that I believe would be an excellent fit with research at IIES. Hereby I will describe four of them.

My *primary area of research* is Monetary Economics. My research agenda within this area would be to contribute to the starting literature of Heterogeneous Agents New Keynesian (HANK) models. Taking this into account, I can't hardly think of a better place than IIES. Starting from the work by Broer et al. (2017)[1], it is interesting to investigate on the distributional effects of monetary[2] and fiscal policy under the New Keynesian (NK) setting. Additionally including myopic agents (in the sense of Gabaix[3]) would help to better build a Behavioural HANK model.

Within the same area, I would like to work with a NK model with habit formation -not only in consumption[4], but also in labor supply- to better match the VAR estimation. In this way one could also account for hump-shape in labor supply impulse response function. It is intuitive to think of individuals supplying a roughly constant amount of labor across time (at least, period by period). A potential problem in this setting is that labor supply would be excessively stable, probably not accounting for fluctuations (interpreted as unemployment in RBC models). Considering this, it would be interesting to additionally include unemployment in the model. Finally, after discussing this topic with my *Monetary Economics* colleagues, I believe it would be interesting to obtain a NKPC that does not depend on output gap but in unemployment gap (as in the ancient curve).

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[1] Tobias Broer, Niels-Jakob Harbo Hansen, Per Krusell and Erik Öberg, The New Keynesian Transmission Mechanism: A Heterogeneous-Agent Perspective, NBER Working Paper No. 22418.

[2] See Greg Kaplan, Benjamin Moll and Giovanni L. Violante, Monetary Policy according to HANK, revised version for American Economic Review, April, 2017

[3] A behavioural New Keynesian model (2017)

[4] Lawrence J. Christiano, Martin Eichenbaum, and Charles L. Evans , "Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy," Journal of Political Economy 113, no. 1 (February 2005): 1-45.

Another idea would be to revisit the hybrid New Keynesian Phillips curve (HNKPC) in a behavioural approach. My idea parts from Gali and Gertler's (1999)[5] research. They estimate the different parameters, under both standard NKPC and HNKPC and find unsatisfactory results: particularly, they find $\theta \simeq 0.86$ (in data $\simeq 0.75$[6]) and $\beta \simeq 0.93$ (in data $\simeq 0.99$) when estimating the NKPC and $\theta \simeq 0.82$ and $\beta \simeq 0.9$ when estimating the HNKPC. In a recent working paper, Gabaix (2017) rewrites the NK model with rational innatention. Particularly, he rewrites the NKPC by including an extra parameter $M \in (0, 1)$ multiplying expected inflation in the next period. Using Gabaix (2016) $\beta M \simeq 0.93$, and given that $\beta \simeq 0.99$, this leads to an attention parameter $M \simeq 0.94$. Unfortunately, this parameter is still away from literature estimates (M usually between 0.85 and 0.9). However, if we consider the HNKPC (which Gabaix does not do), $\beta M \simeq 0.9$ that leads to an attention parameter $M \simeq 0.91$. Microfounding several other NKPC's under rational inattention and estimating them would be the objective of this article.

I would also like to research on the price puzzle[7], both empirically and theoretically. I consider of vital importance to obtain a model that explains such puzzle, and whose inflation impulse response function looks more similar to the empirically observed.

Lastly (within monetary economics), a final idea (very preliminary) would be to include stickiness in the nominal interest rate setting of Central Banks - i.e., including unobservable shocks to them, or shocks that are only observable after one period (Central Bank decision making takes time).

Apart from the aforementioned monetary economics research, the focal point of my present research lies in the literature of Social Security (SS) Reform. Most papers in the literature conclude, based on demographic projections in advanced economies, that moving from Pay-As-You-Go (PAYG) public schemes to Fully Funded (FF) (i.e. privatizing SS) is beneficial for most of individuals. This contention is especially strenuous if one assumes that young individuals (those most benefited by reform) transfer resources to the old.[8]

However, empirical evidence has so far strongly contradicted this view, with the only major economy enacting such a reform (Chile) having clearly underperformed. Quoting Barr and Diamond (2009),[9]

> "[...]pension provision became highly concentrated, calling into question the plausibility of effective competition[...]The number of participating firms is small[...]At 1.76%, concern about the level of charges has been recurrent since the start of the individual accounts".

I believe that the explanation of pension overpricing resides in life expectancy uncertainty. Mortality rates have been repeatedly overestimated in previous studies.[10] My intention is to model this market imperfection (through monopolistic competition), giving rise to profits for firms, and proceed similarly to Kitao (2014).

To resume, the first part of the paper would be to analyze such an environment, and investigate if a change of scheme is desirable. In a second part, I would propose a reform: one in which public SS takes the form of FF for an initial age period[11] and then onwards PAYG,[12] motivating that such a scheme would reduce market imperfections.

Together to this letter, I also enclose copies of my previous work. My master thesis (supervised by Luisa Fuster) investigated on the welfare loss effects of large fees in the Pension sector. In a very simple model (deterministic and with only one asset), I obtained that welfare losses are sizeable. It was at that time, while working on this project, when I was iniciated to Matlab. After that, and thanks to the *Quantitative Macroeconomic Methods* course, I carried my computational skills to a new level. Additionally, I include

[5] Inflation dynamics: A structural econometric analysis, Journal of Monetary Economics 44 (1999) 195-222.

[6] Nakamura, E. and Steinsson, J., Five facts about prices: a reevaluation of menu cost models, Quarterly Journal of Economics, 123(4), 1415-1464, November 2008.

[7] In the sense of Giordani P (2004), An alternative explanation of the price puzzle. Journal Monetary Economics 51(6):12711296

[8] For a clean and nice example of this, see Kitao, S. (2014), Sustainable Social Security: Four Options, Review of Economic Dynamics.

[9] Pension Reform: A Short Guide (2009).

[10] Previous literature has used two life tables: Faber and Wade (1983) and Bell and Miller (2005). For example, Faber and Wade (1983) predicted that the survival rate of 0.5 was at age 76.86 (at age 76.86, there was half chance to die in the next year). Bell and Miller (2005) observed that the realization of such event happened at age 78.20.

[11] e.g. from age 65 to 75

[12] Beneficial in two aspects: (1) even though a (representative) firm pools all pensioneers (no adverse selection), the firm might not interiorize this and will set prices (or interest rates paid on annuities) to obtain excessive profit margins (hence, in such case, having a FF scheme might induce firms to set excessive charges like in Chile), and (2) a PAYG scheme for the *very* elder would be less costly (covers less periods) and would solve the adverse selection problem.

a copy of the term paper (coauthored with Max Viskanic[13] and Mario Luca[14]) for Tore Ellingsen's course *Institutional and Organizational Economics*, on a topic that I consider interesting (belief transmission), although irrelevant for my macroeconomics research (I just enclose it so that you can have a term paper). Unfortunately, none of the macro courses that I took were evaluated in that way. To that purpose, I look forward to take the *Labor-Market Macroeconomics, The Macroeconomy in the Long Run* and (if it happens) the HANK courses next year.

I appreciate you for taking the time to review my application. Awaiting eagerly to work with you.

Yours faithfully,

Jose E. Gallegos

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